

Generative AI-Based Skincare Recommendation System
— LLM + RAG Architecture (Ateeb's Module)

User Layer

User Input
(Skin Image + Symptoms
+ Hormonal Profile)

Firebase Auth
(User Profile & History)

React.js Frontend
(Symptom Form + Image Upload)

API Layer

FastAPI Backend Gateway (Request Routing, Input Validation, Auth Middleware)

Vision AI Module
(CNN / ViT)
Skin Condition Detection
[Team Module]

RAG Pipeline
(LangChain)
Query Embedding
+ Vector Retrieval

Knowledge Base
(FAISS Vector Store)
Dermatology Papers
+ Clinical Guidelines

Sentence Transformers
(Embedding Model)

Ateeb's Module

Prompt Builder
(Structures condition,
hormonal data &
retrieved chunks)

LLM Recommendation Engine (Claude / GPT-4 API)
Prompt: Condition Label + Hormonal Profile + Retrieved Knowledge Chunks
Output: Skincare Routine + Ingredient Advice + Diet Plan + Follow-up

GenAI Layer

Skincare Routine
(AM/PM Steps
+ Product Picks)

Ingredient Advice
(Niacinamide, Retinoids,
AHAs — with warnings)

Diet Plan
(Nutrient-specific
recommendations)

Follow-up
Monitoring
Schedule

Output Layer

LLM-Powered Chatbot (LangChain Conversational Memory)
Handles follow-up user queries with full conversation context — grounded in same RAG knowledge base